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# Room temperature electron spin coherence in photogenerated molecular spin qubit candidates

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**Abstract.** One of the main challenges in the emerging field of molecular spintronics is the identification of new spin qubit materials for quantum information applications. In this regard, recent work has shown that photoexcited chromophore–radical systems are promising candidates to expand our repertoire of suitable candidate molecules. Here, we investigate a series of three chromophore–radical compounds composed of a perylene diimide (PDI) chromophore and a stable 2,2,6,6-tetramethyl-piperidin-1-oxyl (TEMPO) radical by transient electron paramagnetic resonance (EPR) techniques. We explore the influence of isotope labelling of the TEMPO moiety on the EPR spectra and electron spin coherence times of the molecular quartet states generated upon photoexcitation and illustrate that (i) a coherent manipulation of the spin state is possible in these systems even at room temperature and that (ii) a spin coherence time of  $0.7\ \mu\text{s}$  can be achieved under these conditions. This demonstration of electron spin coherence at ambient temperatures paves the way for practical applications of such systems in functional molecular devices.

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## 1 Introduction

A lot of recent work in the emerging field of molecular spintronics has focussed on the identification of novel materials that can generate, transfer, and store spin information efficiently.<sup>[1–5]</sup> Among these materials, photoexcited chromophore–radical systems have emerged as a new class of compounds with promising properties.<sup>[6,7]</sup> In perylene diimide (PDI)–radical derivatives, it could, for instance, be shown that the molecular quartet state formed after photoexcitation has properties required for quantum logic and memory applications and is thus viable as a spin qubit:<sup>[8,9]</sup> in analogy to  $\text{NV}^-$  centres, it can be initialised by light in a highly pure initial state. In addition, it is long-lived, and exhibits coherence times that are much longer than typical pulse microwave-driven gate operation times, even at moderate temperatures of 80 K.

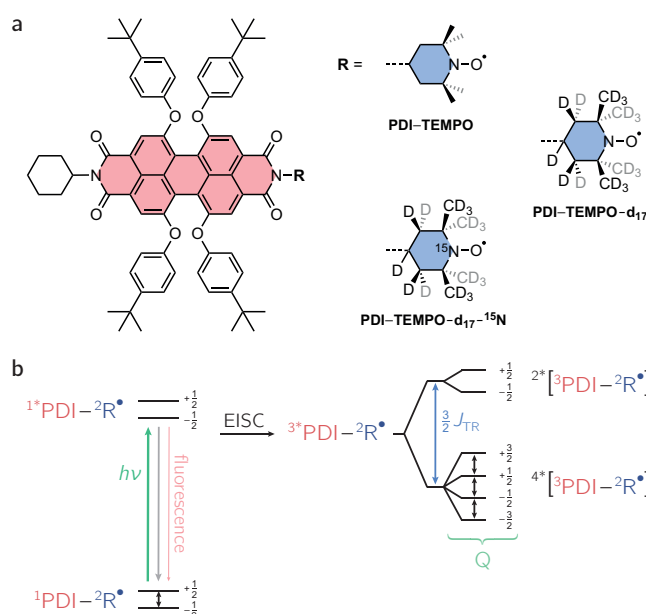
This demonstration of coherent spin properties in frozen solution at 80 K was remarkable since most previous studies on molecular qubit candidates were limited to liquid helium temperatures, often as low as 2–10 K.<sup>[10–12]</sup> However, for the integration of such coherent molecular spin systems in practical spintronic devices, it would be desirable to be able to measure them at even higher temperatures, ideally under ambient conditions.

The magnetic properties of some chromophore–radical systems investigated thus far have already been studied by transient continuous wave electron paramagnetic resonance (trEPR) spectroscopy in liquid solution,<sup>[9,13,14]</sup> but a coherent manipulation of the resulting spin states, by pulse EPR spectroscopy at room temperature, has not been demonstrated, yet.

In the present study, we investigate a series of three different PDI based chromophore–radical systems by trEPR and transient pulse EPR techniques. The three molecules differ with respect to the isotope labelling of the TEMPO (2,2,6,6-tetramethyl-piperidin-1-oxyl) radical moiety. The optical properties of the unlabelled

PDI–TEMPO system were already studied earlier,<sup>[8,15]</sup> allowing us here to focus on the effect of isotope labelling and different measurement conditions on the coherent spin properties of PDI–TEMPO.

## 2 Results and Discussion



**EPR characterisation in frozen solution.** In our previous study,<sup>[8]</sup> we reported on the magnetic properties of unlabelled PDI–TEMPO in isotropic frozen solution at 80 K. Transient nutation measurements revealed that all features in the transient spectrum can be attributed to a species with quartet multiplicity. Further, we showed

that the complex central feature, observed in the trEPR spectrum of PDI–TEMPO, can be reproduced in spectral simulations if the effect of hyperfine coupling to the nitrogen nucleus of the TEMPO moiety is taken into account in addition to the zero-field splitting of the quartet state. The four quartet sublevels are thus further split by the nitrogen ( $I = 1$ ) hyperfine interaction, resulting in a total of twelve accessible spin states. In addition, pulse EPR measurements showed that this photogenerated molecular multi-level system may be viable as a spin qudit candidate as it can be initialised by light in a pure quartet spin state, is long-lived and exhibits coherence times that are longer than typical pulse microwave-driven gate operation times.

Spin qudits (i.e., qubits with more than two levels) can only be realised with organic<sup>[8,9]</sup> or inorganic<sup>[10–12,18–20]</sup> molecular high-spin ( $S, I > 1/2$ ) systems or dopants (as opposed to mesoscopic solid state systems such as semiconductors and vacancy centres) and their use in quantum information science has the striking advantages that the number of iterations in quantum algorithms as well as the error rate can, in principle, be decreased and more complex gate operations can be performed in a single physical unit without the need for inter-qubit communication.<sup>[3,21]</sup>

Given the potential of PDI–TEMPO as a photogenerated molecular spin qudit, the aim of the EPR investigations presented here was to explore different possibilities of enhancing the spin coherence times in this molecule and to make the system amenable to a coherent manipulation of the spin state populations under ambient conditions. The two different strategies pursued here to limit spin decoherence are (i) isotope labelling of the radical moiety and (ii) the embedding of the molecules in a rigid polymer matrix.

To explore the effect of isotope labelling of the radical moiety, we started by characterising the three different PDI–TEMPO compounds and identifying any differences between them with respect to their magnetic properties, using trEPR. The measurements were carried out at Q-band frequencies in isotropic frozen toluene solution at 80 K and the corresponding data are shown in Figure 2 together with the best numerical fits to the data.

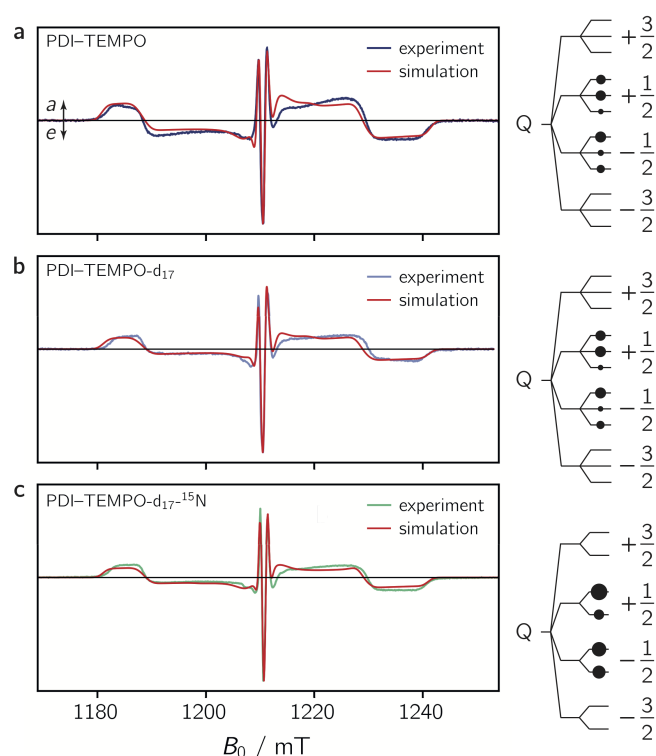


Figure 2: TrEPR spectra of the three dyads recorded at the Q-band in isotropic frozen toluene solution at 80 K together with the best numerical fits to the data. The spectra were recorded in direct detection mode ( $a$ : enhanced absorption,  $e$ : emission). To the right of each spectrum, the relative quartet (Q) sublevel populations are visualised by circles of different sizes. The simulation parameters are given in the text and a detailed overview is provided in the SI (Tables S1–S3).

Comparing the experimental transient EPR spectra of the three dyads (see also Figures S7 and S8 in the SI),

we observe that the overall spectral shape is qualitatively similar in all cases. All spectra are characterised by an *aeaeae* multiplet polarisation (*a*: enhanced absorption, *e*: emission) and a dominant central feature, the so-called "net polarisation" associated with the quartet  $|m_S = +\frac{1}{2}\rangle \leftrightarrow |m_S = -\frac{1}{2}\rangle$  transition, with *aea* polarisation. While the central feature was previously proposed to arise from an unequal population of hyperfine states,<sup>[8]</sup> the observed multiplet polarisation pattern indicates a selective population of the high-field quartet sublevels with  $|\pm \frac{1}{2}\rangle$ , which was suggested to be caused by a mechanism involving electron transfer from TEMPO to PDI followed by fast charge recombination to the chromophore triplet state.<sup>[15]</sup>

Regarding the multiplet polarisation of the coupled spin system, we would not expect any qualitative difference between the three dyads since no difference in the triplet state formation mechanism is expected. However, given that the central feature was assumed to arise from a splitting of the quartet sublevels by the nitrogen hyperfine interaction of TEMPO, the observation of a central three-line pattern for all compounds including PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$  was unexpected at first, but could be clarified with the help of numerical simulations of the spectra using EasySpin<sup>[22]</sup> (v6.0.0-dev.50) as will be explained in the following. A detailed overview of the simulation strategy is presented in the SI (Section 2.5) and a visual explanation of the observation of a central three-line pattern for both the PDI–TEMPO and PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$  compounds is provided in Figure S14.

For the simulation of the trEPR spectra, a coupled triplet–radical spin system ( $S_1 = 1$ ,  $S_2 = \frac{1}{2}$ ) was assumed. The number of variable simulation parameters could be significantly reduced by a separate determination of the coupling parameters of the triplet and radical precursors, which were then kept fixed during the fitting procedure. To obtain the **g**- and **A**-tensors of the radical, a global fit of a X-band continuous wave (cw) EPR spectrum and a Q-band pulse EPR spectrum in frozen solution was performed (Figures S3 and S4), while  $g_T$ ,  $D_T$ , and  $E_T$  of the triplet state were obtained from a simulation of the trEPR spectrum of PDI (see Figure S6). The only variable parameters were thus the coupling parameters ( $J_{TR}$ ,  $D_{TR}$ ) as well as the populations of the spin states. The latter are visualised next to the trEPR spectra in Figure 2.

The peak separation and linewidth of the central spectral feature is different for PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$ , which correlates well with the expected difference in hyperfine coupling. The simulations reveal further that the shape of the complex central feature is not only affected by the populations of the hyperfine levels, but also by the coupling regime (i.e., the exchange and dipolar coupling parameters). We find that, in particular, the outer spectral features and spectral intensities can only be reproduced well, if the system is assumed to be in the upper intermediate coupling regime with  $|J_{TR}| \sim 1200$  MHz. This coupling regime is still consistent with the observation of quartet signatures in transient nutation experiments<sup>[8]</sup> and might, at the same time, explain the unequal population of the hyperfine states, as the quartet and excited doublet states are close in energy and mixing between them is efficient. Such an unequal population would be rather unexpected for a system in the strong coupling regime since the energy difference between the hyperfine levels is very small compared to that between the quartet and excited doublet states.

In summary, the magnetic parameters and sublevel populations were found to be very similar for unlabelled PDI–TEMPO and PDI–TEMPO- $d_{17}$ , which is consistent with the fact that the spectral shape is mainly dominated by the zero-field splitting and nitrogen hyperfine interactions. A careful analysis of the trEPR spectra of all three dyads by spectral simulations revealed that the observation of a central three-line pattern for all isotopologues can be ascribed to the system being in the intermediate coupling regime, where additional transitions are expected (see also Figure S14).<sup>[7]</sup> However, the difference in peak intensities and peak separations between PDI–TEMPO and PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$  can only be explained if the contribution of nitrogen hyperfine coupling of TEMPO to the spectral shape is taken into account.

**Coherence decay in frozen solution.** To explore the influence of isotope labelling on the spin coherence times of both the nitroxide radical as well as the quartet spin state generated upon photoexcitation, pulse EPR measurements were carried out at the Q-band in frozen toluene solution at 80 K. The spin coherence times ( $T_m$ ) measured at a field position corresponding to the intensity maximum of the nitroxide signal (see Figure S5 for the pulse EPR

spectra) are compared for the three PDI–TEMPO dyads in Figure 3a. To obtain the  $T_m$  values listed in the figure, the data were fitted with a single stretched exponential of the form<sup>[23–25]</sup>

$$V_{SE} = \exp\left(-(2\tau/T_m)^\beta\right) \quad (1)$$

where  $\tau$  indicates the inter-pulse delay in the Hahn-echo detection sequence. While the trace for unlabelled PDI–TEMPO shows no discernible nuclear ESEEM effects, clear nuclear modulations with a frequency characteristic for deuterium are superimposed on the decay of the two  $d_{17}$ -labelled isotopologues. The variations in the measured coherence times are relatively minor, ranging from  $3.8\ \mu\text{s}$  for unlabelled PDI–TEMPO to  $4.4\ \mu\text{s}$  for PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$ . Deuteration of the protons on TEMPO seems to have only a very minor influence, as reflected by the similarity of the decay measured for PDI–TEMPO and PDI–TEMPO- $d_{17}$  ( $4.0\ \mu\text{s}$  vs.  $3.8\ \mu\text{s}$ ), while the effect of the replacement of  $^{14}\text{N}$  by  $^{15}\text{N}$  is somewhat larger ( $4.0\ \mu\text{s}$  vs.  $4.4\ \mu\text{s}$ ).

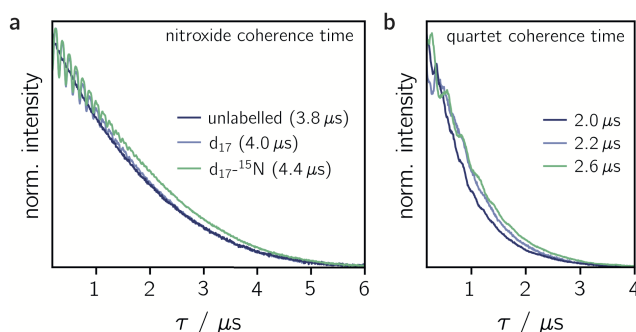


Figure 3: Comparison of the spin coherence times of the nitroxide (a) and quartet state (b) for the three different PDI–TEMPO dyads. The values obtained by fitting of the data using a stretched exponential function are indicated in the figure. The nitroxide coherence times were measured at the intensity maximum of the corresponding field-swept echo-detected EPR spectrum, corresponding to  $B_0$  values of 1208.4 mT, 1208.2 mT, and 1208.7 mT for PDI–TEMPO, PDI–TEMPO- $d_{17}$ , and PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$ , respectively. The coherence times of the corresponding quartet states were measured at a field position corresponding to the central emissive line, i.e.  $B_0$  values of 1210.5 mT, 1210.3 mT, and 1210.5 mT. All magnetic field values refer to a microwave frequency of exactly 34.0 GHz.

The fact that the effect of isotope labelling on the coherence times of the nitroxide is small, is in line with literature observations suggesting that the rotation of the methyl groups is the dominant factor causing spin decoherence in TEMPO radicals. For instance, a replacement of the two methyl groups on each side of the NO-group by spirocyclohexyl has been shown to lead to much improved coherence times over a wide temperature range.<sup>[26,27]</sup>

Figure 3b shows the coherence times measured for the quartet state  $|m_S = +\frac{1}{2}\rangle \leftrightarrow |m_S = -\frac{1}{2}\rangle$  transition at a field position corresponding to the central line with emissive spin polarisation. The values obtained for  $T_m$  are again listed in the figure. Interestingly, the effect on  $T_m$  observed for the labelled nitroxide radicals is perfectly reflected in the corresponding quartet state coherence times. It seems that the coherence time of the radical has a direct impact on the coherence time of the quartet state generated by photoinduced spin mixing: Between PDI–TEMPO and PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$  an increase in  $T_m$  from 2.0 to  $2.6\ \mu\text{s}$  (factor 1.3) is observed, which compares well with the increase of  $T_m$  by a factor of 1.2 observed for the corresponding radicals. This observation could turn out to be crucial for any future investigations and applications of photogenerated high-spin systems, as it suggests that the coherence properties of the photogenerated state can be optimised and further enhanced by improvement of the radical coherence times, for which synthetic strategies already exist.<sup>[28]</sup>

**Room temperature spin coherence.** To demonstrate electron spin coherence under ambient conditions, we developed experimental protocols to embed the molecules in rigid polymer films suitable for pulse EPR measurements at Q-band frequencies. Details regarding the sample preparation using poly(methyl methacrylate), PMMA, can be found in the SI (Section 2.2). Since PDI–TEMPO- $d_{17}$ - $^{15}\text{N}$  showed the most promising magnetic properties in frozen solution, room temperature pulse EPR experiments are only shown here for this compound, while the

corresponding data obtained for unlabelled PDI–TEMPO are presented in the SI (see Figures S10 and S11).

While no transient pulse EPR measurements are feasible with these molecules in isotropic solution at room temperature, the partial immobilisation of the molecular motions by insertion of the compounds in a polymer host matrix makes them amenable to a coherent spin manipulation by microwave pulses, as illustrated in Figure 4. Figure 4a shows the echo-detected field-swept pulse EPR spectra of the nitroxide and the light-induced quartet state. To obtain the spectrum of the pure quartet state, a small residual nitroxide signal needed to be subtracted from the data. While the nitroxide spectrum is remarkably similar to that measured in isotropic frozen solution (see Figure S5), the intensity of the outer wings of the quartet spectrum (multiplet polarisation) relative to that of the central features (net polarisation) is found to be considerably reduced in polymer films at room temperature (see also Figure S10). The central quartet feature with *aea* polarisation, which dominates the appearance of the corresponding frozen solution spectrum, can, however, clearly be recognised and identified. To confirm the assignment of this feature to the quartet  $|m_S = +\frac{1}{2}\rangle \leftrightarrow |m_S = -\frac{1}{2}\rangle$  transition, transient nutation experiments were performed as detailed in the SI and shown in Figure S12. Referenced against the nitroxide signal (pure doublet,  $\omega_0$ ), a frequency of  $2\omega_0$  was determined experimentally, as would be expected based on  $\omega_{m_S, m_S+1} = \omega_0 \cdot \sqrt{S(S+1) - m_S(m_S+1)}$ ,<sup>[29–31]</sup> leaving no doubt about the quartet nature of the observed signal.

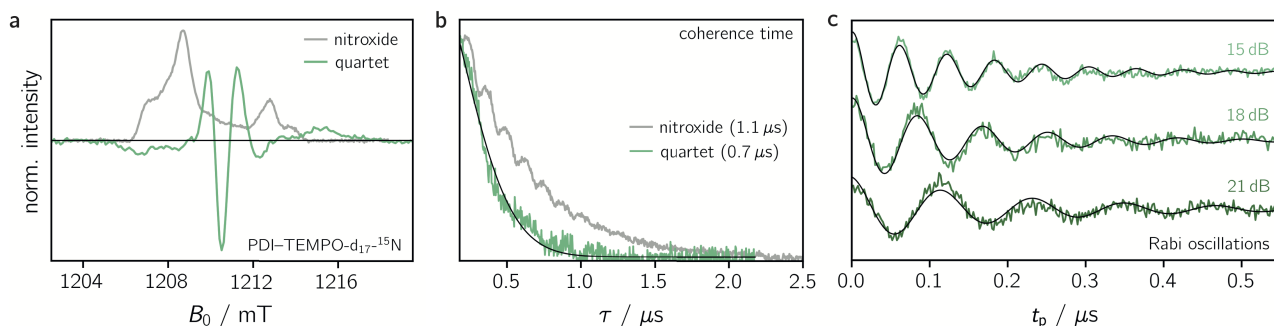


Figure 4: Illustration of the coherent spin properties of the molecular quartet state of PDI–TEMPO- $d_{17-15}N$  in a PMMA host matrix at room temperature. Room temperature Q-band pulse EPR spectra of the nitroxide and quartet state (**a**), spin coherence decays measured for the nitroxide and quartet state (**b**), and Rabi oscillation experiments on the quartet state recorded at different microwave attenuations (**c**). The coherence time of the nitroxide and the coherence time and Rabi oscillations for the quartet state were measured at a magnetic fields of 1208.7 mT and 1210.5 mT, respectively, referring to a microwave frequency of 34.0 GHz.

The measured coherence decays for the nitroxide and quartet state are shown in Figure 4b. A stretched exponential fit to the data yields  $T_m$  values of  $1.1\ \mu s$  and  $0.7\ \mu s$  for the nitroxide and quartet state, respectively. The ratio of the nitroxide and quartet coherence times is found to be similar compared to that in frozen solution ( $4.4\ \mu s/2.6\ \mu s = 1.7$  vs  $1.1\ \mu s/0.7\ \mu s = 1.6$ ), supporting our previous statement that these coherence times might be directly correlated.

Although the coherence time of the quartet state is reduced by a factor of about four at room temperature compared to the measurements at 80 K, the measured  $T_m$  value of  $0.7\ \mu s$  at 295 K still compares favourably with values reported for other molecular spin qubit candidate systems investigated under ambient conditions.<sup>[16,17]</sup> In addition, the investigated system represents the only photogenerated molecular system studied so far under these conditions and we expect that an optimisation of the film preparation conditions (molecular packing) and/or the choice of a different polymer host material may lead to a further enhancement of  $T_m$ .

To demonstrate the possibility of a coherent manipulation of the photogenerated spin system, Rabi oscillation experiments were carried out in addition to the measurement of the coherence decay, as shown in Figure 4c. The measurements were performed at different microwave attenuations (as indicated in the figure) to demonstrate that the Rabi frequency is linear with the applied microwave  $B_1$  field. For microwave attenuations of 15 dB, 18 dB, and 21 dB, Rabi frequencies  $\Omega_R$  of 16.2, 12.0, and 8.3 MHz were obtained, respectively, demonstrating the linearity and thus confirming that the oscillations are of pure Rabi type.<sup>[21]</sup> Considering the measured values for  $T_m$  and  $\Omega_R$ , the number of single-qubit logic operations,  $\Omega_M$ , feasible with this system under the applied conditions, can be

calculated from  $\Omega_M = 2 T_m \Omega_R$  and amounts to  $\sim 22$ . This shows that the quartet state can reach any arbitrary superposition state and can be coherently manipulated for at least 20 rotations at room temperature.

### 3 Conclusions

By investigating a series of three PDI–TEMPO dyads, we explored two different strategies to make photogenerated coupled spin systems amenable to a coherent manipulation of the spin state at room temperature using pulse EPR techniques. The latter requires an enhancement of the spin coherence properties which may either be achieved (i) by isotope labelling to limit the coupling to neighbouring protons or (ii) by embedding the molecules into a rigid matrix to limit decoherence by molecular motion.

Isotope labelling of the TEMPO moiety was found to have only a minor influence on the measured spin coherence times. However, it turned out to be a convenient method to investigate the relationship between the coherence time of the radical precursor and that of the photogenerated high-spin state. It appears that any increase in the nitroxide coherence time is directly correlated with a corresponding increase in the coherence time of the quartet state, generated by spin coupling. Since synthetic strategies to enhance the coherence times of stable radicals already exist, this observation may be crucial: We expect that it will provide an important handle for improving the design of photogenerated multi-spin systems and their applications in QIS.

In addition, isotope labelling could be used here to unambiguously confirm the contribution of hyperfine coupling to the shape of the EPR spectrum measured for the coupled system, which was suggested in our earlier work.<sup>[8]</sup> Numerical simulations of the EPR spectra, using a spin Hamiltonian approach, further reveal, that the coupling between the triplet and doublet spin centres, generated after photoexcitation, falls in the upper intermediate coupling regime. Consequently, we could show that this system represents one of the rare examples<sup>[32]</sup> of covalently-linked photogenerated high-spin systems where it is possible to determine the magnitude of the exchange coupling constant  $J_{TR}$  reliably.

Importantly, we demonstrated here that room temperature EPR measurements are well feasible with photogenerated organic multi-spin systems when embedding them in rigid polymer matrices and we expect that the experimental protocols developed in this context will also be transferable to other photogenerated high-spin systems, making them amenable to pulse microwave manipulations at room temperature. To our knowledge, the present study also represents the first observation of room temperature quantum coherence in a photogenerated organic triplet–doublet spin system which exhibits properties that are suitable for applications in the field of quantum information technology,<sup>[5,7,8,33]</sup> such as quantum sensing. Coherence times close to 1  $\mu$ s and Rabi oscillations could be measured under ambient conditions, which is a first and necessary step for the integration of such or similar systems in practical molecular spintronic devices.

**Supporting Information.** Synthetic procedures and characterisation, UV-vis spectra, description of the EPR sample preparation, EPR measurement conditions and parameters, complementary EPR data, details of the data analysis including simulation parameters.

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**Author contributions.** Spectroscopic characterisation of the compounds, EPR data acquisition and analysis, draft editing and data visualisation M.M.; Establishment of the methodology for the preparation of polymer film samples suitable for Q-band pulse EPR measurements at room temperature, EPR sample preparation, draft editing P.T.;



EPR data analysis and simulations, draft editing T.Q.; Synthesis and characterisation of the compounds, draft editing A.V.J.; Conceptualisation, project administration, funding acquisition, supervision of the optical and EPR experiments, original draft writing, data visualisation S.R.

**Competing interests.** The authors declare no competing interests.

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